

Exploring **Project Gardener**

Juan Diego · STACKIT interview

9 July 2026

WHAT I BUILT

- ☒ One cloud VM (EC2)
- 👓 KinD-based local setup
- 📦 1 Garden + 1 Seed + 1 Shoot
- 🔗 Deployed via `gardener-operator`



GARDEN → SEED → SHOOT

GARDEN

The Brain: Orchestration & State

SEED

The Host: Runs Control Planes

SHOOT

The Cluster: End-user workload

Garden: manages the state.

Seed: a cluster that hosts the control planes of other clusters (hypervisor-style, but the guests are control planes running as pods).

Shoot: the actual cluster delivered to the customer.

KEY COMPONENTS

API Server

Extends K8s with Shoot, Seed, and Project resources.

Scheduler

Intelligently matches new Shoots to the best available Seed.

Gardenlet

The "Pull" Agent: Seed dials out to Garden for instructions.

MCM

Machine Controller Manager: Handles VM lifecycle per provider.

KUBECEPTION: PODS AS CONTROL PLANES

SEED Cluster ns: **shoot--local--local**

kube-apiserver

etcd

kube-scheduler

controller-mgr

🧠 Control Plane = **PODS** (Shared/Cheap)

↓ manages via VPN tunnel ↓

SHOOT Worker Node (Real VM)

User Pod

User Pod

...

Why it matters:

❌ **Normal K8s:** Needs dedicated VMs for control-plane nodes.

✅ **Gardener:** Control plane is just a set of pods. One Seed can host **hundreds** of clusters.

KUBECEPTION: THE TRADE-OFF

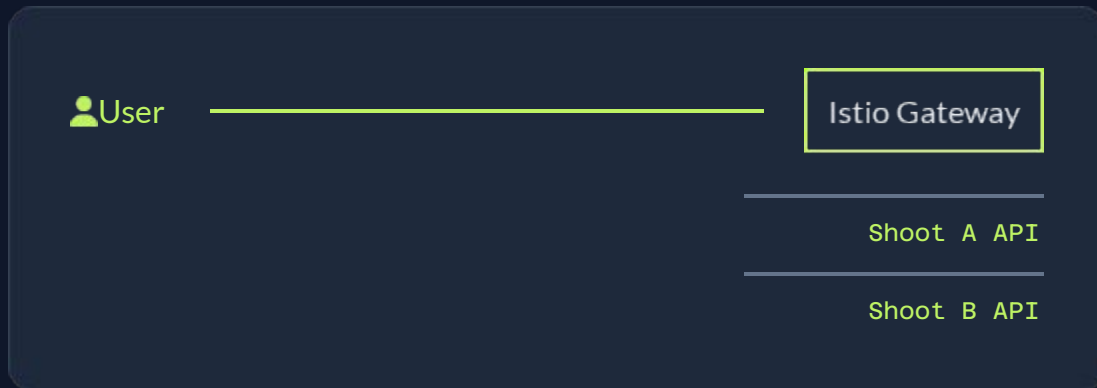
✓ Benefits

- Massive cost reduction (no Master VMs).
- Strict isolation: Control plane is physically separate from customer nodes.

! Challenges

- **Noisy Neighbors:** Control planes share Seed resources.
- Mitigated via resource limits/quotas and NetworkPolicies.

NETWORKING: ONE GATEWAY (SNI)

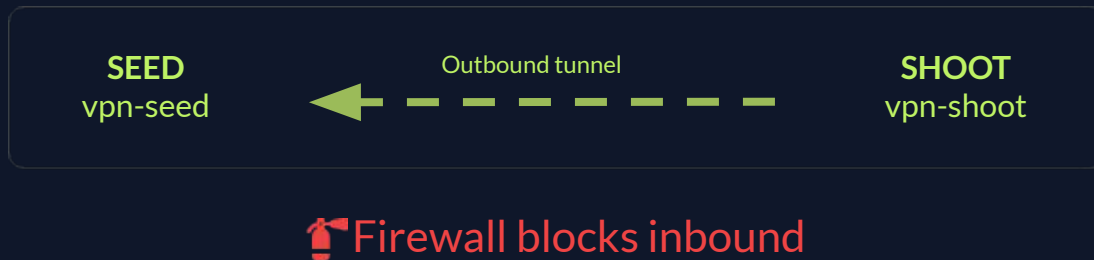


Routing by Hostname

Istio reads the **SNI field** in the TLS handshake.

Routes traffic **without decrypting** it. End-to-end encryption is preserved.

NETWORKING: REVERSED VPN



Outbound-only: the worker opens the tunnel

Workers are often behind strict firewalls.

Solution: The worker starts the connection outward to the Seed and opens the tunnel — the control plane then reaches back through it. Who opens the connection isn't the same as which way traffic flows

THE VERDICT

YES.

For STACKIT / Private Cloud

- ✓ **Proven at scale:** SAP & T-Systems use it for thousands of clusters.
- ✓ **Efficient:** The hosted control-plane model is a game changer for margins.
- ✓ **Extensible:** Swappable extensions for any infrastructure.

RISKS: THE "AIR-GAPPED" CHALLENGE

Images

Mirroring & syncing hundreds of containers into internal registries is a massive chore.

Updates

Fast release cycle requires a robust manual process for offline environments.

Internet Assumptions

Gardener assumes public DNS and Cert-Managers. Providing internal equivalents at scale is where the real engineering effort lies.

WHAT WORKED & WHAT DIDN'T

Status: Healthy

Full Garden/Seed/Shoot stack successfully deployed on a single VM.

Debugging Battle-log

- Docker network clashes (KinD).
- HTTP vs HTTPS registry errors.
- Shoot applied against the wrong API endpoint.

"Breaking things is how I truly understood the architecture."

Live Demo

Questions & Deep Dive

➤_Ready to show the clusters in action.